

Alastair Lee (李瀚琛)

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EDUCATION

Certification of Completion of Coding Bootcamp

Aug 2022 – Nov 2022

University of Toronto, Toronto, ON

Master of Science in Chemical Engineering

2020 – 2021

University of Rochester, Rochester, NY

Bachelor of Science in Chemical Engineering

2016 – 2020

University of Rochester, Rochester, NY

WORK EXPERIENCE

Mindful Sparks, Hong Kong

Brewery Operator

Jan 2025 - Present

- Operated an ISO certified microbrewery bottling machinery for a sparking tea brewery that can process over two thousand bottles of different sizes a day
- Passed an internal examination for daily cleaning of the bottling machinery following the CIP (Clean in Place) protocol.
- Labelled and packaged bottles for distribution both locally and to international distributors

University of Rochester, Rochester, NY

Research Assistant, Computational Fluid Dynamics Group

Aug 2019 – May 2021

- Analyzed fluid flow in 30 patient-specific carotid arteries to identify patterns that could indicate stroke using ANSYS Fluent
- Discovered trends in the velocity data of stroke patients that suggest a notable difference between stroke and non-stroke patients
- Improved procedure for analysis by discovering shortcuts in the program that reduced the amount of manual processing time by up to 80% while maintaining the same accuracy
- Functioned as a supervisor and mentor to other members of the group, helping to correct mistakes and answer questions on how to further analyze each patient

Teaching Assistant, Lab in CHE Principles

Aug 2020 – Dec 2020

- Managed five groups of three students each in their Lab projects, advising them and helping to keep track of their progress
- Attended biweekly meetings to critique each groups' pre-lab reports and extrapolate their potential difficulties they would have in lab
- Assisted students outside of assigned hours by answering questions or listening to presentation rehearsals through Zoom

PROJECT EXPERIENCE

University of Rochester, Rochester, NY

Co-Chairman and Former Business Manager, Chem E Car Club

Aug 2018 – May 2020

- Led a 15-person team in creating a miniature vehicle for an annual competition hosted by the American Institute of Chemical Engineers
- Created a battery to function as the power source for the vehicle using a magnesium dioxide cathode and a zinc anode
- Established a production line to create the battery, utilizing each team members' expertise to create prototypes most efficient for rigorous testing
- Designed and built the chassis of the vehicle using CAD and 3D printing

- Organized weekly briefing sessions and created timelines to coordinate workflow between team members, reallocate resources as needed, and ensure critical deadlines were achieved
- Functioned as Business Manager in the 2018-2019 school year, ordering chemicals and battery materials, and interfacing with the Department of Chemical Engineering

Student Researcher, Medical Center

Jan 2020 – May 2020

- Investigated polymers to create kidney models for medical students' training
- Led group testing for how reordering the formulation procedure would affect the Young's Modulus and clarity of the resulting polymer, using a factorial DOE
- Detailed a technical transfer report to convey findings and experiments completed

Electoral Board Member, American Institute of Chemical Engineers

Chapter

Aug 2016 – May 2020

- Co-led a team to participate in the annual Pumpkin Launch Competition, winning the event back-to-back years despite fewer team members and less expertise compared to other engineering organizations
- Acted as intermediary between Chapter President and board member to assist with work allocation
- Participated in weekly meetings to organize and publicize events for the student body, including research talks and social events

Student Researcher, Lab in CHE Principles

Aug 2019 – Dec 2019

- Led and designed group project in testing pump system with both vane pump and centrifugal pump
- Constructed a Piping and Instrumentation Diagram of the system for wrap-up presentation and report
- Completed group project using gas chromatography, analyzing how differences in injection technique, time, and cleanliness can affect the results of the GC test

OTHER SKILLS

Web Development:

JavaScript ES6+, CSS3, HTML5, SQL, NoSQL, MongoDB, MySQL, Apollo GraphQL, Express, Node, Handlebars, jQuery, Bootstrap, Materialize, Python 3

Lab Skills:

Mathematica, LabVIEW, AutoCAD, Space Claim, Design Modeler, and NX

NMR Analysis, IR Analysis, Gas Chromatography, Mass Spectroscopy, and Titration

Spoken Languages:

English, Cantonese, and Mandarin